

Nome: Rafael Duarte Pereira da Silva

Nº: 1190979

Data: 16/04/2026

Título Provisório: Human-centric intelligent environments: From signals to adaptive environments

Orientador: Prof. Luis Gomes

Co-orientador: Prof. Zita Vale

Problem Statement:

The field of artificial intelligence is rapidly transforming various sectors, and its application in intelligent buildings presents a significant opportunity to enhance human-centric adaptation [1], [2]. One of the core challenges is to create systems that are adaptive to human needs, interactive in a natural way, and ensuring privacy and responsiveness [3].

In user modeling, general solutions focus more on behavioral modeling, using approaches such as deterministic, stochastic, or agent-based [4]. However, for preference modeling, more traditional solutions rely on rule-based methods [5], that while providing many advantages, such as explainability, do not capture nuanced variations in human preferences or handle the complexity in multi-occupant scenarios [6]. The emergence of graph-based representations [7], [8], [9], particularly knowledge graphs, has provided a more robust method for modeling the connections between users, devices, spaces, and contexts, when compared with machine learning solutions which typically capture behavioral patterns from sensor data [10].

From a scientific standpoint, the organic modelling of user preferences, representing all the complex connections between contents and user, has been a challenge that was not yet covered in a standard way [5], [11]. However, without the correct modelling of users, in what way personal assistants and AI-based solutions could really connect to users? This issue is not a new issue, but with the growth of AI, is becoming, more and more, significant, being a key aspect in the AI-user symbiotic relation [12].

Interaction between human and buildings has evolved from simple dashboards to voice-activated commands. While an improvement, these forms of interaction often lack context-awareness, and are largely transactional and reactive [13]. The integration of large language models (LLM) [14], [15], [16] and small language models (SLM) [17], [18] present an opportunity to create fluid interactions and, also, enable proactive assistance. One opportunity is to integrate these language

models with near-real-time and structured system context so they can leverage proactive preference elicitation, personalized system behavior, and handle conflict mediation [19], [20].

In conclusion, while the literature presents advances in user preference modeling, distributed systems, and language models, there is a gap in research that combines these concepts into a unified framework. The work will evolve from the model proposed by the candidate in [21], and [22]. This project directly addresses this multi-faceted gap, aiming to deliver a holistic organic solution that is more adaptive, interactive, and resilient than the current state of the art.

Research Questions:

Main RQ. How can intelligent systems continuously synthesize and evolve holistic occupant representations to proactively infer latent intent and fulfill the heterogeneous, dynamic needs of individual and collective occupants?

RQ1. How can a dynamic and organic user preference profile be generated and evolve from user's behavioral, environmental contexts, and interactions?

RQ2. How can intelligent systems proactively engage with human users based on the recognition of latent intent from real-time signals and inferred preferences?

RQ3. How can multiple holistic and organic user preference profiles be interconnected to accommodate active learning and negotiations in a shared context?

RQ4. How can an intelligent human-centric community be boosted by the proposed holistic and organic user preference modelling?

Objectives:

O1. Analyze and synthesize existing approaches to occupant modeling, preference inference, conflict resolution, and privacy-preserving multi-agent architectures in intelligent buildings.

O2. Evaluate the suitability of heterogeneous relational representations for dynamic occupant profiling, analyzing their capacity to integrate multimodal data streams and support downstream preference inference.

O3. Investigate and assess mechanisms for inferring occupant preferences and contextual intent from dynamic profiles, analyzing how inference accuracy evolves across varying conditions and temporal patterns.

O4. Investigate the role of language models in proactive occupant engagement, evaluating their capacity to recognize latent intent and mediate preference elicitation and negotiation through natural dialogue.

O5. Analyze conflict patterns arising from competing occupant preferences in shared spaces and compare resolution strategies against fairness, satisfaction, and individual autonomy criteria.

O6. Assess the viability of privacy-preserving peer-to-peer resource and knowledge sharing between building agents, evaluating the trade-off between community-level contextual exchange and individual occupant privacy.

O7. Evaluate the integrated framework across Digital Twin simulation and real-environment pilots, assessing each component's contribution to the system's capacity to continuously anticipate and fulfill individual and collective occupant needs.